PROJECT REPORT

CPE Studio AI
Features & technology

A simple guide to what has been built, how each feature works, and the tools behind it.

1. Check devices
2. Join & record
3. Enhance & transcribe
4. Download

Recording and studio features

Device setup & testing

Tool: Browser device APIs + Web Audio API

Choose a detected internal or external camera, microphone and speaker. Test the camera, play a speaker tone, or record your voice until you choose **Stop and listen**. Includes a live microphone meter and headphone setting.

Recording controls

Tool: MediaRecorder + Web Audio API + IndexedDB

A **5-second countdown** comes before recording. Start, pause/resume and stop/save controls are available. Screen sharing is optional and can begin after recording starts. Camera and microphone can be turned on or off.

Studio layout & controls

Tool: HTML, CSS and JavaScript

Dynamic camera, side-by-side, slides focus, picture-in-picture and slides-only previews. Side-by-side places **slides left and camera right**. Includes the People panel toggle, device arrows, hover labels and a compact Effects button.

Review & downloads

Tool: FFmpeg + Python ZIP tools

Download **camera + mic MP4**, **muted slides MP4** when shared, and **transcript TXT** individually or together in a ZIP. Exports use a shared 30 fps timeline for equal video lengths. Camera preview and export use the mirror view.

How it runs: start the Python project from Terminal and use its interface at localhost in a desktop browser. It is a local project, not a packaged desktop app.

Current scope: a single-host prototype. Layouts change the preview; downloads remain separate tracks. The Recordings library has been removed, so download from the review page.

AI AFTER RECORDING

Enhance, then transcribe

Magic Audio and Eye Contact have separate switches above the Transcript section. Their status changes from Applying to Applied; Eye Contact shows No adjustment when no gaze change is made.

Magic Audio

Tool: FFmpeg arnndn filter + RNNoise-style RNN model (std.rnnn)

The model estimates background noise in the recorded microphone audio. The application blends cleaned audio with the original and keeps its timing aligned with the video. Switching it on processes the **full camera recording** for review and export.

Current scope: noise reduction. A separate audio loudness normalization control and a full audio mastering chain have not been added.

Eye Contact - beta

Tool: MediaPipe Face Landmarker + custom Python/NumPy adjustment + FFmpeg

Face and iris points are located in each frame. Small pixel shifts around the eyes gently move the apparent gaze toward the camera, while smoothing changes between frames.

Current scope: a conservative eye adjustment, not a trained gaze-redirection model. It works only on suitable, nearly front-facing frames and may leave a recording unchanged.

Speech-to-text transcript

Tool: Whisper base through MLX Whisper or Faster Whisper

Speech is converted into readable paragraphs. The transcript can be viewed below **Generate transcript** and downloaded as TXT **without visible timestamps**. It currently uses the original recorded microphone audio.

Where it runs: locally after the model is downloaded. MLX Whisper uses the Apple Silicon GPU; Faster Whisper uses CPU or a compatible NVIDIA GPU on other supported systems.

Turning both enhancement switches off restores the original camera recording. Processing creates an enhanced version; the original recording is retained.

LIVE CAMERA FEATURES

Framing, backgrounds & appearance

These tools are available before joining the studio. Selected appearance and background effects continue in the studio and are included in the recorded camera track.

Framing guidance & low-light tip

Tool: MediaPipe Face Landmarker + JavaScript brightness checks

A fixed guide turns green for good framing and yellow when you are off-center. It gives positioning advice without moving or cropping the camera. A low-light tip opens the appearance controls.

Background blur & solid colors

Tool: MediaPipe Image Segmenter + Selfie Multiclass model + Canvas

The model separates the presenter from the room. Choose no blur, slight blur or full blur, or replace the room with a solid color. Edge quality can vary around hair, hands and low-light scenes.

Touch-up

Tool: MediaPipe face mask + Canvas blur and blending

Light smoothing is blended into the face region, with eyes, eyebrows and mouth protected from the smoothing mask. Two strength settings control the amount.

Studio lighting

Tool: Canvas brightness filter + JavaScript smoothing

A slider brightens or dims the camera scene with gradual transitions. This is a brightness adjustment, rather than a separate AI relighting model. The Face lighting option has been removed.

Screen glow

Tool: HTML, CSS and JavaScript

An adjustable light border offers brightness and warm/cool controls, with local dimming near the pointer. It lights the display area and stays visible in the studio. The border is not drawn into the camera MP4.

Two different kinds of lighting: Studio lighting changes camera pixels. Screen glow changes the light shown by your display. Framing guidance only advises you on position and lighting.

TECHNOLOGY & NEXT STEPS

The stack, in plain English

Each AI feature uses the tools listed on the previous pages. The common application stack is below.

Interface	**HTML, CSS and vanilla JavaScript** Builds the setup, studio, review and effect controls.
Camera, microphone & screen	**Browser Media APIs + MediaRecorder + Web Audio API** Finds devices, captures tracks, tests audio and records the session.
Live camera processing	**Web Workers + MediaPipe WebAssembly + OffscreenCanvas** Runs face/background analysis and draws effects without sending camera frames to a cloud service.
Local backend	**Python 3.12 + built-in HTTP server** Serves localhost and handles transcription, enhancements and downloads.
Recording storage	**Browser IndexedDB** Stores recording chunks locally so the session can be saved, reviewed and exported.
Video & file export	**FFmpeg via imageio-ffmpeg + libx264/AAC + Python zipfile** Creates MP4 video with audio, aligns track lengths and packages downloads.

What can be improved next

Magic Audio: compare original and cleaned lecture samples, especially quiet words and sentence endings. Consider a separate loudness normalization option.

Eye Contact: evaluate a trained gaze-redirection model and check natural blinks, glasses and head movement. Add an easy original/enhanced comparison.

Background effects: refine hair and hand edges and check consistency across external cameras and difficult lighting.

Status: this report describes the current implementation, not a quality benchmark. The improvements above are future work. End-to-end camera, audio and export testing remains part of prototype validation.

Prepared from the current local project code on 4 November 2026. Models and dependencies require an initial download; recording, transcription and enhancement processing are designed to run locally afterward.